

Sensitivity-aware inspection prioritisation for archaeological landscapes: a reproducible Taxila observatory

Abstract

Conservation teams need to prioritise inspections across archaeological landscapes using environmental observations collected at different spatial and temporal scales. We present the Climate-contextual Heritage Inspection Prioritisation (CHIP) framework and evaluate how its recommendations change under alternative indicator, spatial-support and uncertainty assumptions at the Taxila World Heritage property, Pakistan. The framework combines 55 Landsat scenes across five epochs, approximately 30 m terrain data, daily weather for 1991–2025 and 17 geolocated heritage components. A hierarchical landscape–terrain score is evaluated through domain and factor ablation, three neighbourhood radii, 27 structural scenarios, exact weight-transition analysis, spatial-block resampling, coordinate perturbation and joint spatial–decision simulation. Of 386,280 common-support cells, 16.93% met at least two adverse spectral criteria. Removing MNDWI changed rank concordance to $\rho = -0.280$, whereas removing NDVI or NDBI retained $\rho > 0.989$, showing that information content was unevenly distributed across indicators. Giri led the fixed 500 m ranking across landscape weights of 0.105–0.769 and entered the top three in 73.24% of joint score vectors; its 95% rank interval nevertheless spanned 1–9. Spatial reconstruction and decision assumptions produced distinct inspection rationales, particularly for Bhallar. Opposing precipitation trends in Open-Meteo and NASA POWER limited climatic attribution. In a separate seven-family spatial land-cover comparison, development-selected random forest achieved macro-F1 0.821 (95% interval 0.691–0.835); feature controls and calibration experiments exposed limitations that a single accuracy score would conceal. CHIP provides a reproducible inspection portfolio linking component priorities to environmental drivers and decision assumptions, with independent monument-condition validation remaining a subsequent field task.

Keywords: cultural heritage; Taxila; remote sensing; spatial uncertainty; inspection prioritisation; reproducibility.

1 Introduction

Archaeological conservation requires decisions across spatial scales: monument fabric, its immediate surroundings and the wider landscape. Satellite observations, terrain models and weather products capture different parts of this system. Translating them into an inspection programme requires more than overlaying indicators: the resulting priorities must be traceable to their environmental drivers and stable enough for a defined management task.

Remote sensing and GIS have documented development pressure and supported heritage multi-hazard assessment [1–3]. At Taxila, Khan et al. integrated ground-based and satellite evidence for hazard assessment [4]; Butt et al. subsequently examined land-cover transitions and urban encroachment over 1990–2024 [5]. These studies establish the local conservation problem. Recent work by Megarry et al. extends the relevance of surrounding land cover to climate-vulnerability assessment across cultural World Heritage properties in the Indian Subcontinent [6]. The next decision-oriented question is whether an inspection set persists when spatial support, indicator composition and decision weights change. Mapping pressure and evaluating the stability of a management recommendation are complementary contributions.

GIS multicriteria methods formalise indicator aggregation, while sensitivity analysis reveals dependence on weights and spatial representation [7, 8]. Rank-acceptability methods further describe which alternatives remain preferred under uncertain criteria and preferences [9]. For archaeological landscapes, these ideas motivate an experimental treatment of priority construction: correlated proxies can duplicate information, neighbouring buffers can share observations, and uncertain coordinates can alter the sampled environment. A single final ranking leaves these dependencies unresolved.

Climate interpretation introduces temporal-support constraints. Exposure depends on materials, maintenance, drainage and microenvironment alongside meteorological forcing [10–12]. Acquisition-matched weather and cross-product comparison can test whether a proposed explanation is consistent with the observations. Separately, predictive evaluation must distinguish geographic land-cover transfer from conservation-condition assessment. Spatially nearby samples can share predictors and labels [13–15]; reproducing a global land-cover product does not establish monument deterioration.

We develop CHIP as an experimentally evaluated inspection framework with three contributions. First, a hierarchical landscape–terrain rule exposes indicator influence through baselines and ablations, and exact weight crossings identify where inspection leadership changes. Second, multiscale, shared-block, positional and nested spatial–decision experiments distinguish persistent priorities from representation-dependent ones. Third, acquisition-matched climate comparisons and a seven-family buffered land-cover benchmark evaluate the environmental and predictive evidence supporting interpretation. The resulting inspection portfolio retains both a persistent component set and the mechanisms that require field verification.

2 Research aim

The study identifies which Taxila components merit early field inspection and which parts of that recommendation depend on analytical choices. It tests adverse spectral convergence on common spatial support, changes in climate interpretation across acquisition windows and products, rank stability across neighbourhoods and uncertainty designs, and land-cover proxy agreement under buffered spatial evaluation. Mathematical definitions connect these experiments to reproducible component-level decisions.

3 Materials and methods

3.1 Study area, inventory and evidence sources

Taxila is a serial World Heritage property in Punjab, Pakistan, inscribed under reference 139 [16]. The inventory contained 18 component records. Seventeen had usable point coordinates; Saraikala (139-002) remained in the documentary inventory but was excluded from coordinate-dependent analysis because its location was unresolved. Spectral neighbourhoods used WGS 84/UTM zone 43N (EPSG:32643); terrain extraction used local metric approximations on the geographic elevation grid. Point-centred circles of 250, 500 and 1,000 m defined analytical neighbourhoods, with 500 m used for primary reporting. They are neither official property polygons nor statutory or UNESCO buffer zones (Fig. 1).

Landsat Collection 2 Level-2 surface reflectance supplied five post-monsoon three-year composites: 2003–2005, 2008–2010, 2013–2015, 2018–2020 and 2023–2025, abbreviated E2004 to E2024. They contained 11, 10, 8, 8 and 18 retained scenes, respectively. Landsat 7 supplied the first two epochs, Landsat 8 the middle two, and Landsat 8/9 the final epoch. Source records preserve acquisition dates, sensor identities and quality information. Cross-sensor and masking limitations were considered explicitly [17, 18].

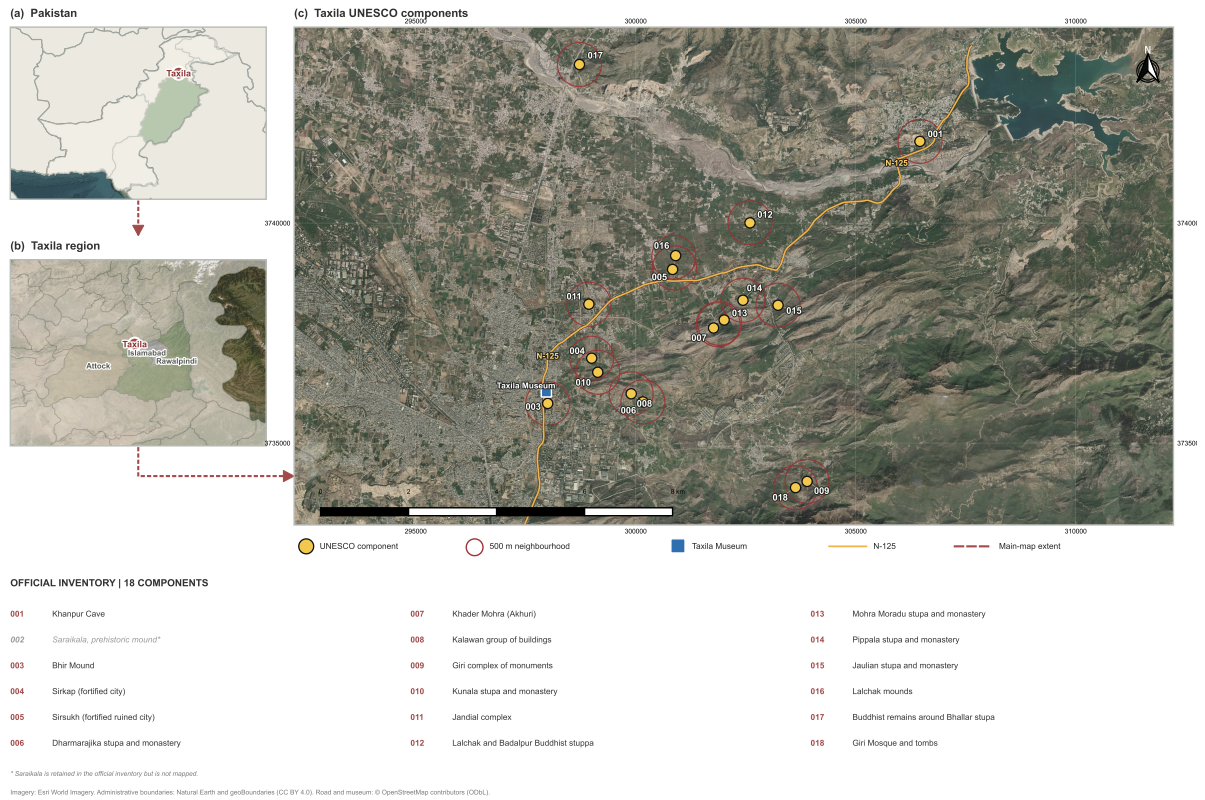


Figure 1: Taxila study-area context and the official 18-component inventory. Seventeen components are mapped; Saraikala is retained in the inventory without an inferred coordinate. Dark-red circles indicate 500 m analytical neighbourhoods. Esri World Imagery provides contextual imagery only; it was not used to calculate spectral, climate or terrain indicators. Boundaries: Natural Earth and geoBoundaries; road and museum: OpenStreetMap contributors (ODbL). Map lines delineate study areas and do not necessarily depict accepted national boundaries.

Table 1: Evidence sources and permitted analytical roles.

Source	Support	Analytical role
Landsat C2 L2	55 scenes; five epochs; 30 m grid	Relative spectral pressure and endpoint convergence
Elevation tile	Approximately 30 m	Terrain and drainage susceptibility
Open-Meteo	Daily, 1991–2025; four cells	Regional trends and acquisition-window context
NASA POWER	Daily, 1991–2025	Weather-product sensitivity
WorldCover consensus	2020/2021 labels matched to E2019 features	Weakly supervised land-cover benchmark
UNESCO component inventory	18 records; 17 mapped points	Component identifiers and neighbourhood centres

Daily Open-Meteo ERA5-Seamless data covered 1991–2025 [19]. Component queries resolved to four unique weather cells, each containing 12,784 daily records. NASA POWER supplied a comparison product over the same period [20]. Neither was treated as station truth. Terrain came from the one-arc-second N33E072 elevation tile distributed through AWS Terrain Tiles. ESA WorldCover 2020/2021 consensus [21] supplied six proxy land-cover classes for an E2019 feature table. Table 1 summarises the role of each source. Figure 2 separates the paths through which these inputs inform interpretation.

3.2 Spectral transformations and common-support convergence

Cloud, shadow and invalid observations were masked before compositing on a fixed 30 m grid. With scaled reflectance ρ_b in band b , the indices were

$$\text{NDVI} = \frac{\rho_{\text{NIR}} - \rho_{\text{red}}}{\rho_{\text{NIR}} + \rho_{\text{red}}}, \quad \text{NDBI} = \frac{\rho_{\text{SWIR1}} - \rho_{\text{NIR}}}{\rho_{\text{SWIR1}} + \rho_{\text{NIR}}}, \quad \text{MNDWI} = \frac{\rho_{\text{green}} - \rho_{\text{SWIR1}}}{\rho_{\text{green}} + \rho_{\text{SWIR1}}}. \quad (1)$$

These ratios follow their original definitions [22–24]. NDBI responds to bare soil as well as built surfaces; its adverse direction is a screening assumption, not an attribution of urban expansion.

For component i , epoch e , radius r and factor k , let x_{ierk} denote the neighbourhood median. The oriented empirical percentile is

$$z_{ierk} = \frac{\text{rank}_{\mathcal{I}_r}(s_k x_{ierk}) - \frac{1}{2}}{N_r}, \quad s_N = s_W = -1, \quad s_B = +1. \quad (2)$$

Ranks ascend with average ties; $N_r = |\mathcal{I}_r| = 85$ component–epoch observations at each radius (17×5). Terrain percentiles use the 17 static component summaries. Spatial reconstruction experiments instead rerank the 17 simultaneously reconstructed E2024 summaries. Reconstruction differences therefore include a reference-population effect. Percentiles describe relative evidence within their stated population, not transferable physical thresholds.

Endpoint differences $\Delta I_k(p) = I_{k,\text{E2024}}(p) - I_{k,\text{E2004}}(p)$ used only the common valid-cell set Ω . With $Q_k(q)$ denoting its empirical q -quantile,

$$C(p) = \mathbf{1}[\Delta I_N(p) \leq Q_N(.2)] + \mathbf{1}[\Delta I_W(p) \leq Q_W(.2)] + \mathbf{1}[\Delta I_B(p) \geq Q_B(.8)],$$

$$\pi_{\geq h} = |\Omega|^{-1} \sum_{p \in \Omega} \mathbf{1}[C(p) \geq h], \quad h \in \{2, 3\}. \quad (3)$$

The indicator function $\mathbf{1}[\cdot]$ equals one when its condition holds. These overlapping, data-derived tails quantify descriptive convergence; they are neither independent tests nor measurements of damaged area.

CHIP From geospatial evidence to inspection decisions

17 geolocated components | 250 / 500 / 1,000 m analytical neighbourhoods

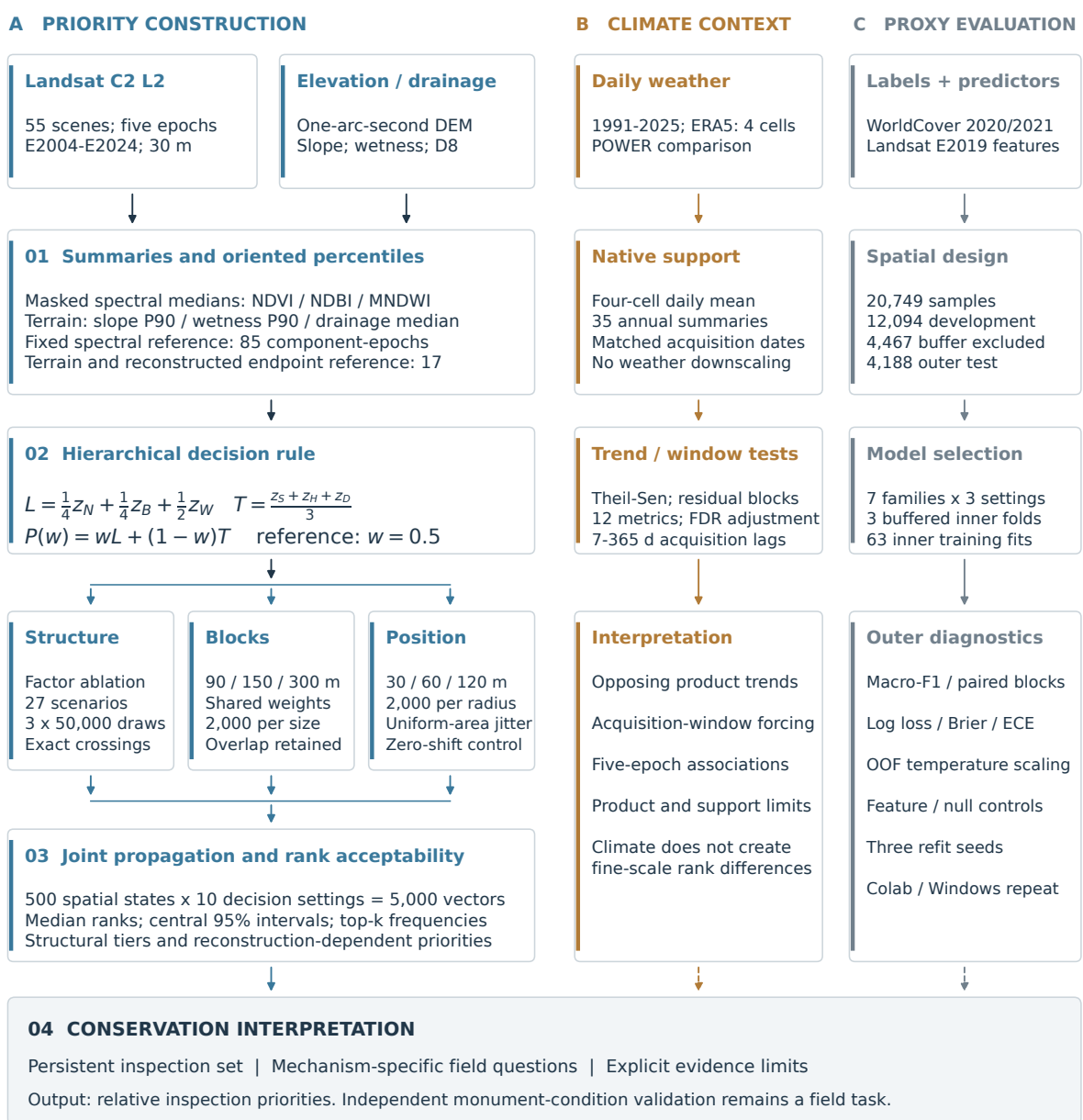


Figure 2: CHIP framework linking observation support, mathematical transformations, experiments and conservation interpretation. Solid arrows carry analytical inputs; dashed arrows carry interpretation. Climate retains native support and land-cover evaluation remains a separate diagnostic. Fixed and reconstructed spectral percentile populations are distinguished. Counts describe executed designs. Codex assisted this editable vector schematic.

3.3 Terrain representation and hierarchical decision rule

D8 routing directs each elevation cell to its steepest strictly lower neighbour; sinks remain terminal. Slope θ derives from metric elevation gradients. The contributing-cell count A_c and geometric-mean cell width h define the numerical wetness proxy

$$H(p) = \log \left[\frac{(A_c(p) + 1) h / (1 \text{ m})}{\tan(\max\{\theta(p), 0.05^\circ\}) + 0.01} \right]. \quad (4)$$

This regularised topographic proxy follows the contributing-area/slope rationale of hydrological terrain analysis [25, 26]. Drainage comprises cells at or above the 99th percentile of accumulation. Component factors are the 90th percentiles of slope and H , and median Euclidean drainage distance. Higher slope and wetness and shorter distance are oriented towards greater susceptibility. Routing approximates topographic convergence, not discharge, and omits unresolved walls, culverts and local drains.

With S, H, D indexing slope, wetness and drainage proximity, landscape pressure L , terrain susceptibility T and priority P are

$$\begin{aligned} L_{ier} &= \frac{1}{4}z_{ierN} + \frac{1}{4}z_{ierB} + \frac{1}{2}z_{ierW}, \\ T_{ir} &= \frac{1}{3}(z_{irS} + z_{irH} + z_{irD}), \\ P_{ier}(w) &= wL_{ier} + (1 - w)T_{ir}, \quad P_{ier}^{\text{ref}} = P_{ier}(0.5). \end{aligned} \quad (5)$$

Grouping NDVI and NDBI limits their combined weight relative to MNDWI without claiming statistical independence. Equal domain weights are declared choices, not fitted conservation parameters. This hierarchical construction connects GIS multicriteria assessment [7] to explicitly testable sensitivity assumptions [8, 27]. A regional climate term remains contextual: adding the same value to every component through an equal-weight integrated score is an affine transformation of P^{ref} and cannot alter the within-epoch ordering.

3.4 Structural experiments and exact weight transitions

The hierarchy was compared with an equal six-factor mean, landscape-only and terrain-only baselines. Factor ablations removed one term and renormalised retained weights. Spearman correlation, Kendall concordance, top-five overlap and maximum rank displacement describe ordering changes.

Three radii, three landscape weights (0.25, 0.50, 0.75) and three terrain formulations produce 27 scenarios. Terrain alternatives comprise equal factors, inverse-correlation weights and the reduced slope–drainage mean. For the correlation-adjusted case, $a_k \propto [\sum_\ell |\rho_{k\ell}|]^{-1}$ with $\sum_k a_k = 1$, where ρ is the Spearman matrix of the three terrain factors, including its diagonal. Continuous-weight experiments sample $w \sim \text{Beta}(\alpha, \alpha)$ for $\alpha \in \{0.5, 1, 2\}$, with 50,000 draws each. This evaluates stated preference distributions rather than inferring stakeholders' preferences, consistent with rank-acceptability analysis [9].

An additional deterministic analysis resolves every pairwise score crossing. For E2024 components i, j at fixed radius and equal-factor terrain, solving $P_i(w) = P_j(w)$ gives

$$w_{ij}^* = \frac{T_j - T_i}{(L_i - T_i) - (L_j - T_j)}. \quad (6)$$

Only nonzero denominators and crossings in $(0, 1)$ are retained. Sorting these crossings partitions the weight interval into regions of constant rank order; midpoint evaluation identifies the leader in each region. Boundary ties retain component-inventory order. Figure 5 combines fixed ranks, a three-dimensional rank response and exact leadership intervals. Radii remain discrete; no interpolated spatial-resolution surface is implied.

3.5 Spatial, positional and joint uncertainty propagation

At each block size (90, 150, 300 m), 2,000 spatial draws used shared exponential weights. If m_{ibk} is the median of factor k in block b intersecting component i , and n_{ib} its valid-cell count,

$$E_b^{(d)} \sim \text{Exp}(1), \quad v_{ib}^{(d)} = n_{ib} E_b^{(d)}, \quad \hat{x}_{ik}^{(d)} = \text{wmed}_b\{m_{ibk}; v_{ib}^{(d)}\}. \quad (7)$$

The weighted median is the first ordered block median reaching half the cumulative weight. Sharing $E_b^{(d)}$ across overlapping neighbourhoods preserves their resampling dependence; cell counts retain unequal spatial support. Terrain remains fixed in this experiment.

For positional sensitivity, independent uniform-area displacements are

$$\delta_i = d_{\max} \sqrt{U_i} (\cos \phi_i, \sin \phi_i), \quad U_i \sim U(0, 1), \quad \phi_i \sim U(0, 2\pi). \quad (8)$$

Spectral and terrain summaries were re-extracted for 2,000 draws at each $d_{\max} \in \{30, 60, 120\}$ m. An undisplaced re-extraction supplies the appropriate control. Equation (2) distinguishes this endpoint reference from the fixed summary-table reference.

Joint simulation generates 500 spatial states, each with sampled block size, shared block weights and independent displacement within 120 m. Ten decision settings per state sample $w \sim U(0, 1)$ and one terrain formulation. For resulting ordinal ranks R_i^{sd} , top- k frequency is

$$\hat{b}_i(k) = \frac{1}{500 \times 10} \sum_{s=1}^{500} \sum_{d=1}^{10} \mathbf{1}[R_i^{sd} \leq k]. \quad (9)$$

Median ranks and central 95% intervals summarise these 5,000 nested score vectors. They are conditional sensitivity distributions, not 5,000 independent maps or posterior probabilities of deterioration. The same frequency definition over 27 structural scenarios defines a core tier at $b_i(3) \geq 0.5$ and a conditional tier outside it at $b_i(5) \geq 0.5$.

3.6 Climate trends and acquisition-matched forcing

The regional daily series averages four unique weather cells equally before annual aggregation. For annual values Y_t , the Theil–Sen estimator is

$$\hat{\beta} = \text{median}_{u>t} \frac{Y_u - Y_t}{u - t}. \quad (10)$$

After robust trend fitting, five-year residual blocks were resampled and the fitted trend restored to obtain 2,000 slope replicates [28, 29]. Null replicates add resampled residuals to a constant baseline. The two-sided probability is $(1 + \sum_b \mathbf{1}[|\beta_b^0| \geq |\hat{\beta}|]) / (2,001)$; Benjamini–Hochberg adjustment covers 12 annual metrics [30]. Inference assumes adequate representation of short-range residual dependence and remains conditional on the weather product.

For acquisition dates $a \in \mathcal{A}_e$, daily precipitation $W(t)$ and antecedent window ℓ ,

$$\overline{W}_e(\ell) = \frac{1}{|\mathcal{A}_e|} \sum_{a \in \mathcal{A}_e} \sum_{u=1}^{\ell} W(a - u), \quad \ell \in \{7, 30, 90, 180, 365\} \text{ days}. \quad (11)$$

Temperature uses window means. Product comparisons match dates and periods. Five composite epochs provide only five temporal replicates for exploratory weather–spectral associations; no weather field was downscaled to 30 m.

3.7 Buffered model selection and probability diagnostics

The weakly supervised benchmark contained 20,749 class-stratified samples, six reflectance bands and four indices. Its outer stripe contained 4,188 samples in 20 blocks; 4,467 samples in adjacent block columns were excluded, leaving 12,094 development samples in 60 blocks. Three contiguous inner validation bands each excluded an adjacent block row from training. This design targets geographical transfer [14, 31], with class-stratified agreement interpreted separately from population-area map accuracy [32, 33].

Seven families—logistic regression, random forest, extra trees, histogram gradient boosting, MLP, XGBoost and CatBoost—each received three predeclared configurations [34–37]. The resulting 63 inner fits selected family m and configuration η through

$$(\hat{m}, \hat{\eta}) = \arg \max_{m, \eta} \frac{1}{3} \sum_{v=1}^3 F_{1, \text{macro}}(y_v, \hat{f}_{m, \eta}^{(-v)}(X_v)),$$

$$F_{1, \text{macro}} = \frac{1}{6} \sum_{c=1}^6 \frac{2\text{TP}_c}{2\text{TP}_c + \text{FP}_c + \text{FN}_c}. \quad (12)$$

Undefined class terms receive zero. Water was absent from one inner training fold; the six-class denominator was retained and unseen-class probability set to zero where necessary. Standardisation was fitted within training folds and early stopping disabled. All families were refitted on development data before outer prediction; selection used only inner scores.

Alongside macro-F1, fixed-six-class mean recall and overall agreement, probability quality used

$$\mathcal{L} = -\frac{1}{n} \sum_{j=1}^n \log p_{j, y_j}, \quad \text{BS} = \frac{1}{n} \sum_{j=1}^n \sum_{c=1}^6 (p_{jc} - \mathbf{1}[y_j = c])^2. \quad (13)$$

Log loss and the unnormalised multiclass Brier score assess probabilistic predictions [38]. Ten equal-width confidence bins define top-label calibration error as $\sum_b (n_b/n) |\text{accuracy}_b - \text{confidence}_b|$. A scalar $\tau \in [0.25, 4]$ minimised development out-of-fold log loss, using

$$p_{jc}^{(\tau)} = \frac{\exp[\log(\max\{p_{jc}, 10^{-12}\})/\tau]}{\sum_{h=1}^6 \exp[\log(\max\{p_{jh}, 10^{-12}\})/\tau]}. \quad (14)$$

The transform follows temperature scaling [39]; its transfer was tested on outer probabilities. Two thousand shared test-block resamples estimated paired macro-F1 differences. Bands-only, indices-only, location-only, shuffled-label and class-prior controls tested information content; three seeds assessed refit sensitivity.

The outer stripe had been examined previously, making this a retrospective comparison. WorldCover consensus labels also differ temporally from E2019 predictors. These experiments therefore evaluate land-cover proxy agreement without establishing independent monument-condition accuracy.

3.8 Computational implementation

Model experiments ran in Google Colab and independently on Windows with recorded package versions, spatial splits, configurations and per-sample probabilities. Derived inputs and deterministic figure/table builders support reproduction [40, 41]. Scene manifests document acquisition and preprocessing. The exact weight analysis uses recorded component summaries without refitting their percentile transforms.

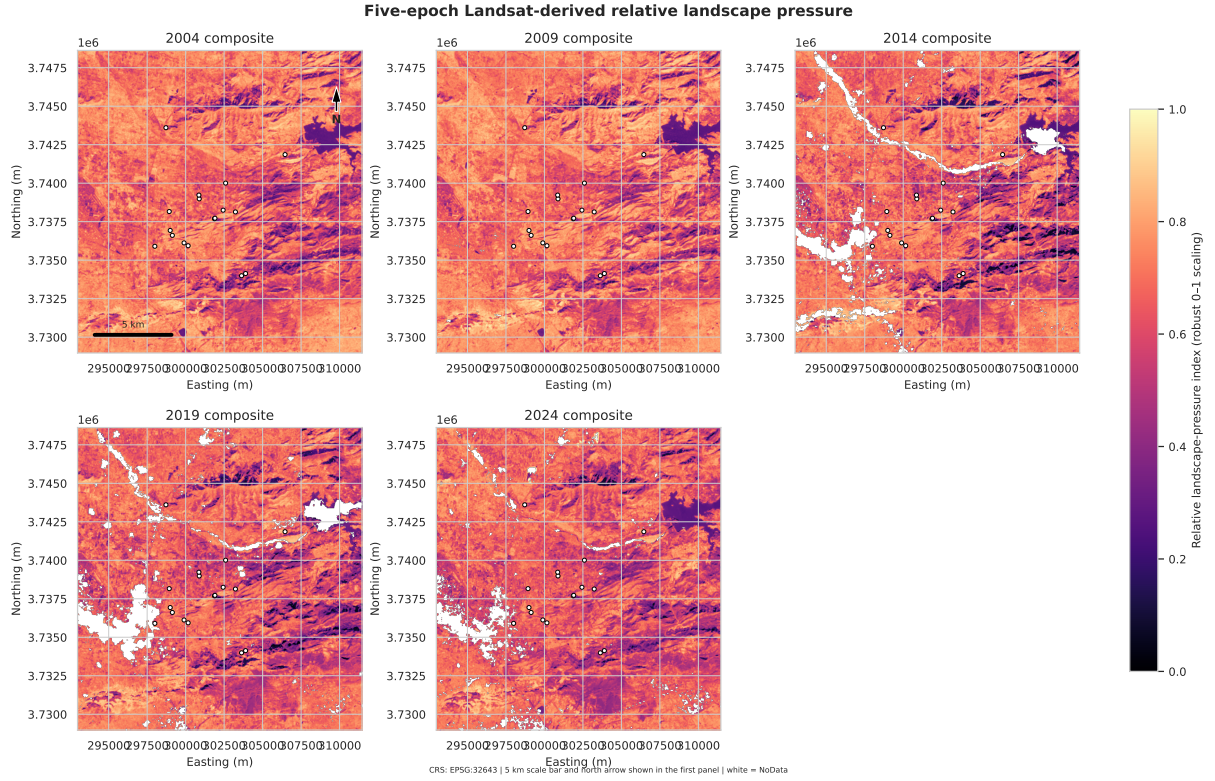


Figure 3: Landscape-pressure patterns across the five Landsat epochs. The common colour scale describes relative pressure; the maps permit comparison of spatial persistence and redistribution across epochs.

4 Results

4.1 Multitemporal landscape pressure and spectral convergence

Landscape-pressure patterns varied across the five epochs (Fig. 3). 386,280 of 403,480 cells (95.74%) were valid at both endpoints, leaving 17,200 cells outside change denominators. On this intersection, 65,391 cells (16.93%) met at least two adverse criteria and 5,493 (1.42%) met all three. The empirical thresholds were $\Delta\text{NDVI} \leq -0.00543$, $\Delta\text{MNDWI} \leq -0.03743$ and $\Delta\text{NDBI} \geq 0.00835$. Figure 4 locates their convergence relative to the wider pressure pattern.

At 500 m, NDVI- and NDBI-pressure ranks were strongly aligned ($\rho = 0.972$). Their grouping therefore affects how surface-cover information contributes to the hierarchy. Giri’s surface-cover term was low (0.0176), while its MNDWI-pressure percentile was 0.9941 and terrain score 0.6765. Its leading composite priority consequently arose from the moisture-related spectral and terrain terms, rather than agreement among every indicator.

4.2 Baselines and ablations identify the drivers of priority

Giri complex led the fixed 500 m score (0.5912), followed by Giri Mosque and tombs (0.5725) and Jaulian (0.5676). The ungrouped equal-factor baseline retained only three of the reference top five, with $\rho = 0.844$ and a maximum movement of seven ranks; it placed Bhallar first. Landscape-only scoring retained one of the top five ($\rho = 0.325$; maximum movement 11) and selected Sirkap, whereas terrain-only scoring retained four ($\rho = 0.830$; maximum movement seven) and selected Bhallar.

Factor ablations were markedly asymmetric (Fig. 6a). Removing NDVI or NDBI retained $\rho = 0.991$ and 0.990 , respectively, and left Giri first. Removing MNDWI instead produced $\rho = -0.280$, no top-five overlap and a maximum movement of 13 positions. Removing slope or

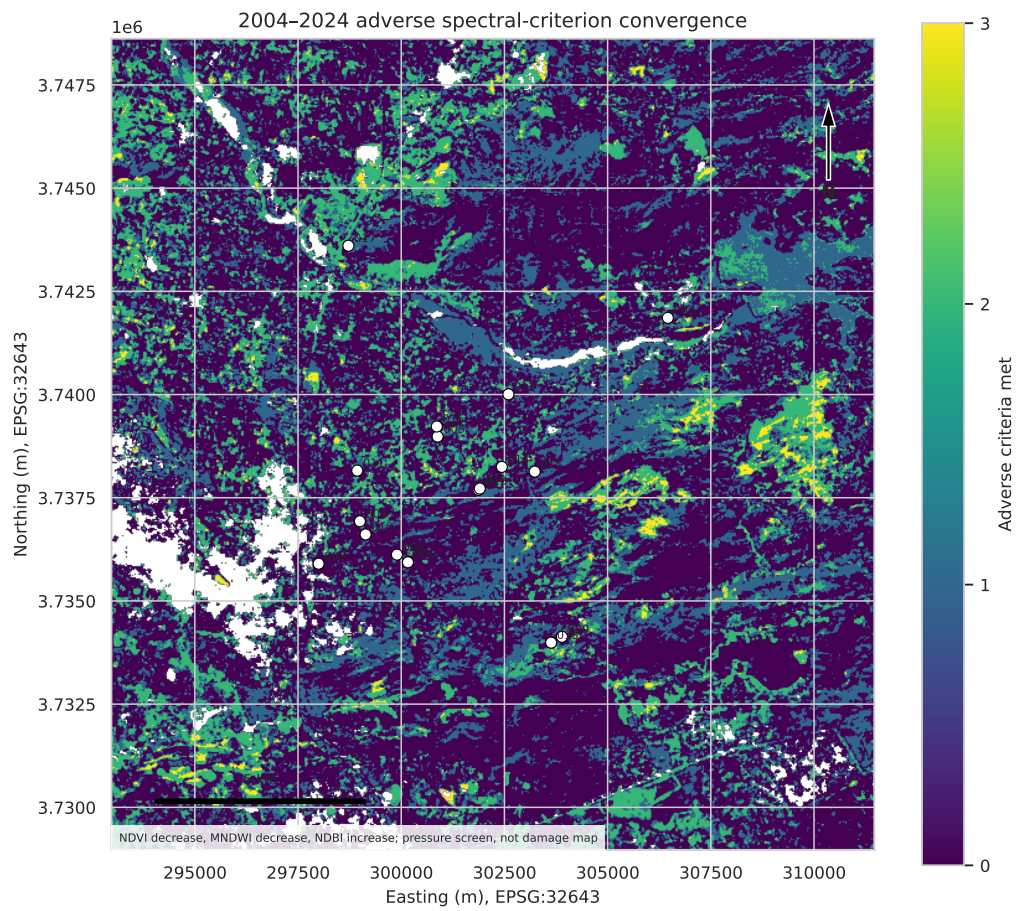


Figure 4: Coincident adverse spectral criteria between E2004 and E2024. Counts use the 386,280-cell common endpoint support. Convergence is a landscape-screening indicator rather than a mapped observation of archaeological damage.

a Fixed component ranks

E2024 | $w = 0.5$ | 17 components

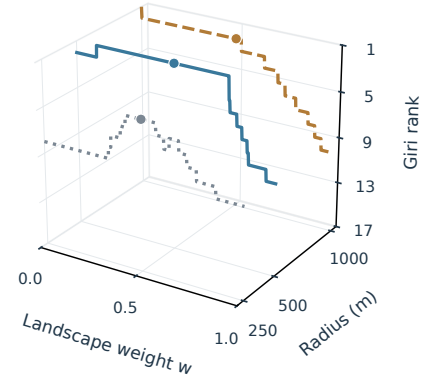
	250 m	500 m	1,000 m
Giri complex	4	1	2
Giri Mosque / tombs	1	2	4
Jaulian	2	3	1
Mohra Moradu	11	4	5
Dharmarajika	3	5	6
Khader Mohra	7	6	3
Kalawan	12	7	9
Bhallar	9	8	7
Sirsukh	8	9	10
Kunala	5	10	13
Pippala	6	11	8
Sirkap	10	12	15
Lalchak mounds	13	13	14
Khanpur Cave	15	14	11
Lalchak / Badalpur	17	15	12
Jandial	16	16	16
Bhir Mound	14	17	17

Top three

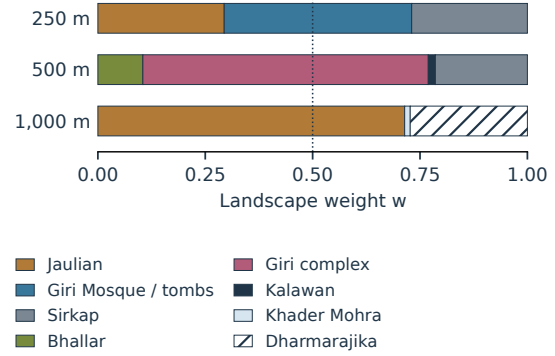
Ranks 4-5

b Weight-by-radius response

Giri complex | discrete radii | dots: $w = 0.5$



c Exact first-priority intervals



Equal-factor terrain; crossings computed analytically.

Figure 5: Scale-weight experiments for fixed E2024 summaries: (a) component ranks at equal domain weights; (b) Giri rank response at three discrete radii (dots: $w = 0.5$); and (c) exact first-priority weight intervals. Terrain factors retain equal weights. The 3D traces depict decision response without interpolating between radii. Supplementary Table S13 reports the leadership intervals.

drainage proximity produced $\rho = 0.611$ and 0.454 and placed Sirsukh first. The hierarchy thus reduced duplication between the vegetation and built/bare terms, while the moisture-related indicator and terrain pathways remained consequential to the decision.

4.3 Spatial support, structural choices and inspection tiers

Rank agreement was stronger between 500 and 1,000 m ($\rho = 0.918$) than between 250 and 1,000 m ($\rho = 0.711$; Fig. 5). Jaulian ranked 2, 3 and 1 at the three radii, whereas Giri complex ranked 4, 1 and 2. Giri Mosque and tombs moved from first at 250 m to fourth at 1,000 m.

Across the 27 structural scenarios, Giri complex, Giri Mosque and tombs, and Jaulian formed the core tier, appearing in the top three in 17, 20 and 18 scenarios, respectively. Khader Mohra and Mohra Moradu formed the conditional tier, appearing in the top five in 15 and 17 scenarios. Scenario agreement therefore supports a small inspection set more clearly than a unique ordering. Continuous domain-weight perturbation reinforced this distinction: Giri's top-three frequency increased from 0.683 to 0.869 as α increased from 0.5 to 2, concentrating weights towards an even balance. Bhallar's frequency decreased from 0.371 to 0.221.

Exact crossings sharpen this interpretation (Eq. 6). At 500 m, Giri leads for $0.1047 < w < 0.7692$: Bhallar leads below this interval, followed at higher weights by a narrow Kalawan

interval and then Sirkap above 0.7857. At 250 m, leadership passes from Jaulian to Giri Mosque and tombs at 0.2941 and to Sirkap at 0.7306. At 1,000 m, Jaulian leads until 0.7143, followed briefly by Khader Mohra and then Dharmarajika. Thus a broad weight interval at one radius does not imply leadership across radii. Exact uniform-weight top-three frequencies agreed with the 50,000-draw experiment within 0.0024 across components.

4.4 Spatial, positional and joint uncertainty

Shared-block resampling gave Giri top-three frequencies of 0.994, 0.971 and 0.960 for 90, 150 and 300 m blocks (Fig. 6b). Bhallar’s frequencies were 0.914, 0.931 and 0.815, while Giri Mosque and tombs declined from 0.773 to 0.390. Larger blocks therefore altered uncertainty unevenly across components.

The undisplaced re-extraction control ranked Bhallar first, Giri second and Jaulian third, demonstrating a difference from the fixed summary-table ordering before positional perturbation. Relative to that control, displacement effects were modest for Bhallar and Giri: at 120 m their top-three frequencies were 1.000 and 0.993. Jaulian declined from 0.915 at 30 m to 0.824 at 120 m, while Giri Mosque and tombs increased from 0.086 to 0.180 (Fig. 6c). Feature reconstruction and positional uncertainty thus make separable contributions and should not be interpreted as the same experiment.

Joint simulation widened the decision envelope (Fig. 6d). Giri had median rank 2, interval 1–9 and top-three frequency 0.7324; Bhallar had median rank 3, interval 1–13 and frequency 0.5260. Giri Mosque and tombs and Jaulian had frequencies 0.4626 and 0.4096. In particular, Bhallar’s prominence under spatial reconstruction contrasted with its limited top-three persistence across the fixed structural scenarios (2/27). This identifies a component requiring a mechanism-led drainage and terrain inspection, rather than treating one aggregate frequency as a universally stable priority.

4.5 Climate trends and acquisition-matched conditions

Open-Meteo annual precipitation decreased by 10.30 mm year^{−1} (95% residual-block interval −17.91 to −2.67; annual-family $q = 0.027$), while annual mean temperature increased by 0.0418 °C year^{−1} (0.0099–0.0731; $q = 0.026$). Heavy-precipitation days declined by 0.143 days year^{−1} ($q = 0.018$). Soil moisture also declined, although these inferences remain conditional on the product and temporal dependence model.

NASA POWER gave opposing point slopes: +29.21 mm year^{−1} for precipitation and −0.0372 °C year^{−1} for temperature. Annual precipitation agreement across 35 years was weak (Spearman $\rho = 0.083$; RMSE 646.74 mm year^{−1}; Fig. 7).

Acquisition-window matching further differentiated the epochs. E2024 antecedent precipitation averaged 42.47, 221.19 and 852.92 mm over 30, 90 and 365 days. The maxima occurred instead in E2004, E2019 and E2014, respectively. Thus the weather preceding the retained satellite scenes cannot be represented reliably by assigning a single wet or dry label to an epoch. Detailed event definitions and annual trends are supplied in Supplementary Table S2.

4.6 Seven-family spatial comparison and diagnostic experiments

Buffered development validation selected random forest with mean macro-F1 0.695. On the 4,188 outer-stripe samples, it achieved macro-F1 0.821 (95% test-block interval 0.691–0.835), agreement 0.813, log loss 0.498 and calibration error 0.014 (Table 2). The MLP achieved the largest outer macro-F1 (0.849), but its lower development score did not support selecting it as the primary model.

Paired spatial resampling showed random forest exceeding logistic regression by 0.0145 macro-F1 (95% interval 0.0028–0.0283). Histogram gradient boosting exceeded random forest by 0.0248

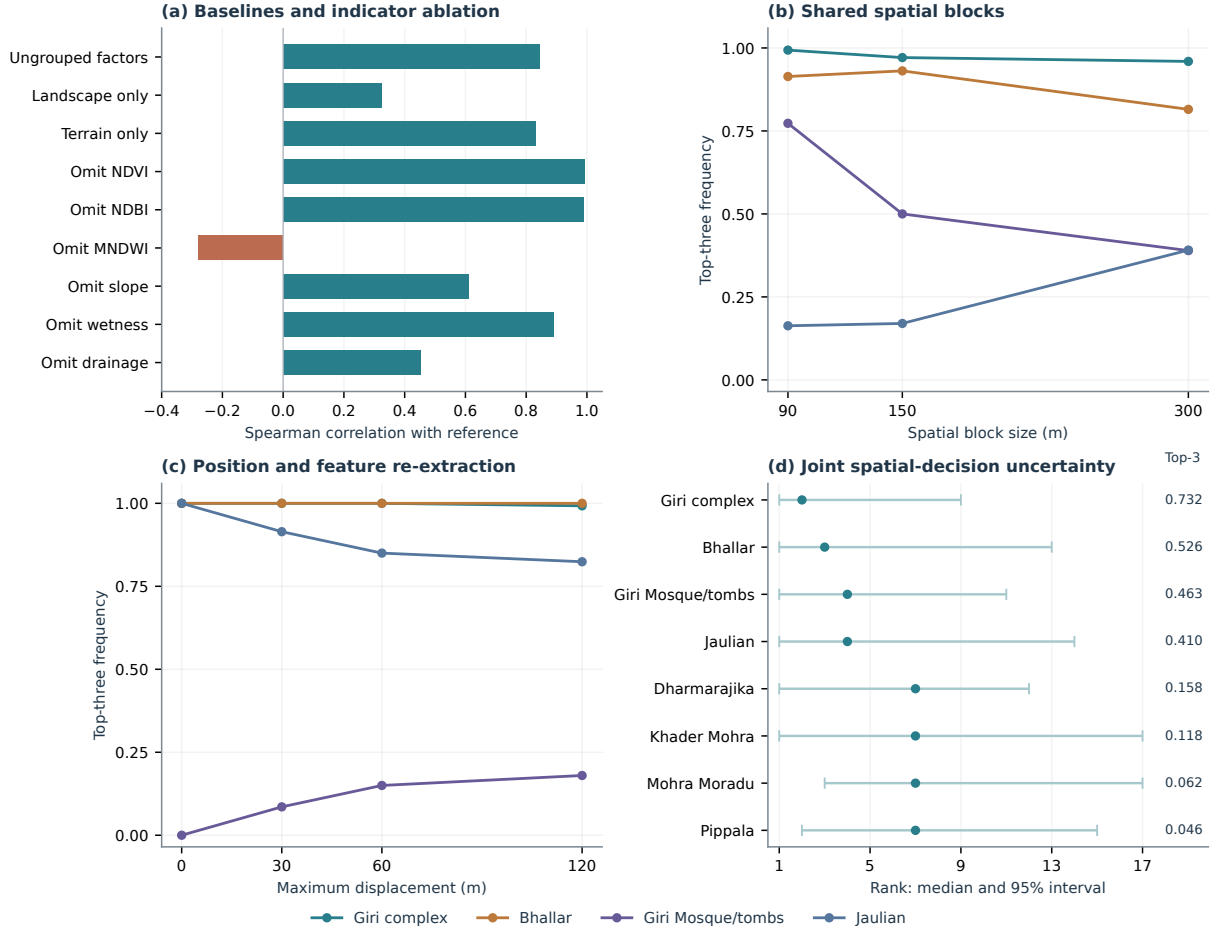


Figure 6: Experimental assessment of the inspection rule: (a) rank concordance after baseline or factor changes; (b) shared-block sensitivity; (c) coordinate-displacement sensitivity with both spectral and terrain re-extraction; and (d) median ranks and central 95% intervals under joint spatial-decision simulation. Labels in (d) give top-three frequencies. Panels (b)–(d) condition on their respective reconstruction and perturbation designs; the zero-displacement control in (c) is distinct from the fixed summary-table reference in (a).

Table 2: Seven-family spatial comparison. The primary model is selected using buffered development folds; outer intervals use 2,000 shared test-block resamples.

Model	Inner F1	Outer F1	95% interval	ECE
Random forest (primary)	0.695	0.821	0.691–0.835	0.014
Logistic regression	0.655	0.807	0.680–0.822	0.050
MLP	0.624	0.849	0.723–0.888	0.020
Extra trees	0.617	0.826	0.694–0.840	0.029
XGBoost	0.616	0.827	0.699–0.845	0.017
CatBoost	0.614	0.792	0.680–0.825	0.017
Histogram boosting	0.611	0.846	0.704–0.857	0.051

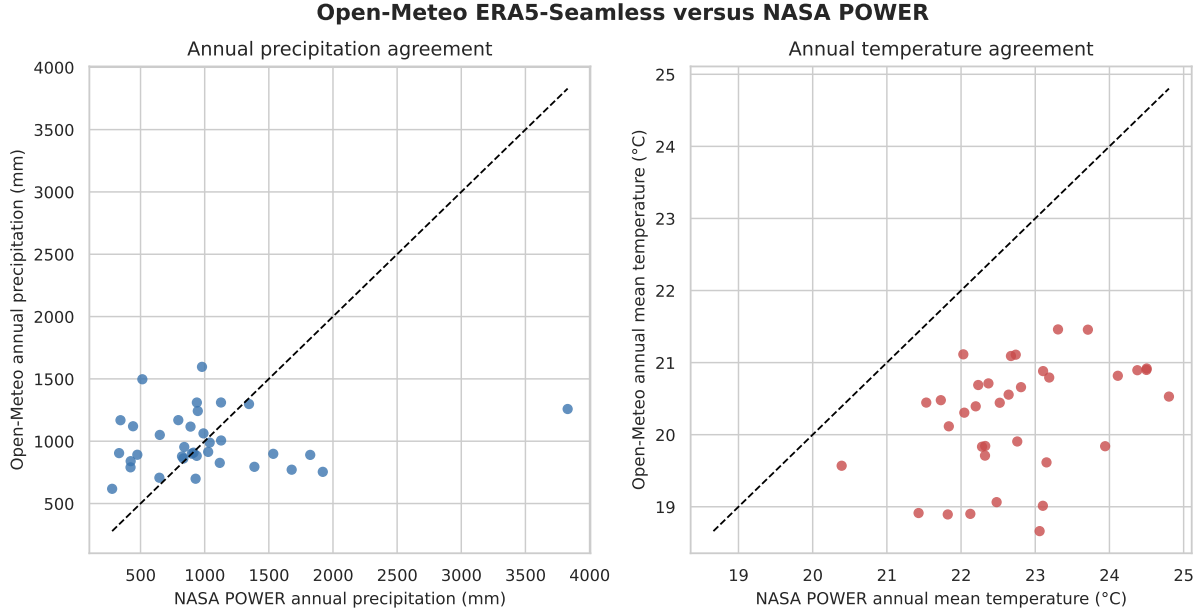


Figure 7: Matched annual precipitation and temperature in the two gridded weather products. Dashed lines indicate equality; product disagreement constrains environmental interpretation in the absence of station validation.

Table 3: Feature controls and probability calibration on the same outer stripe. Feature controls use fixed-capacity logistic regression, except the class-prior control. Temperature is fitted exclusively to development out-of-fold probabilities.

Experiment	Macro-F1	Log loss	ECE
Random forest, raw	0.821	0.498	0.014
Random forest, temperature scaled	0.821	0.574	0.136
Bands only	0.824	0.562	0.051
Indices only	0.671	0.816	0.034
Location only	0.140	1.912	0.147
Shuffled development labels	0.070	1.712	0.063
Class prior	0.049	1.706	0.041

(0.0117–0.0663), whereas the MLP comparison had an interval spanning zero. These unadjusted comparisons distinguish development selection from outer-stripe performance.

The feature controls provide a complementary interpretation (Table 3). Fixed-capacity logistic regression using bands alone reached macro-F1 0.824, whereas indices alone reached 0.671. Location-only, shuffled-label and class-prior controls achieved 0.140, 0.070 and 0.049. Spectral information therefore explained substantially more agreement than coordinates or these null controls, but the selected ten-feature random forest did not outperform the simpler bands-only control by its point estimate.

Development-only temperature scaling ($\tau = 1.658$) increased outer log loss from 0.498 to 0.574, Brier score from 0.269 to 0.300 and calibration error from 0.014 to 0.136. The calibration learned from geographically separated folds transferred poorly to the outer stripe. Random-forest macro-F1 ranged only from 0.8205 to 0.8214 across three refit seeds and matched between Colab and Windows, indicating that the larger differences arose from modelling and spatial-evaluation choices rather than the selected model’s refit seed.

5 Discussion

5.1 From environmental indicators to inspection decisions

The principal contribution is the experimental evaluation of an inspection rule across several distinct sources of uncertainty. A component’s score is accompanied by evidence about its drivers, its persistence across support and structural choices, and its response to spatial reconstruction. Inspection teams can consequently identify where to visit and which mechanism to investigate.

The ablations show why the hierarchy matters. NDVI and NDBI supplied closely aligned information, while MNDWI and terrain pathways strongly affected the ordering. Giri’s leading score was consequently associated with moisture-related spectral and terrain terms despite low surface-cover pressure. An appropriate field visit would investigate drainage routes, surface moisture and their relationship to the monument fabric, while a landscape-led priority such as Sirkap would motivate examination of surrounding surface-cover change.

The results complement Taxila research combining ground observations and satellite evidence [4], and wider heritage multi-hazard assessments [3, 42]. CHIP adds a structured way to test how priorities respond to analytical choices before those priorities are operationalised. Its landscape-context rationale also complements the recent South Asian World Heritage assessment of Megarry et al. [6]; here, that context is translated into component-level decision sensitivity rather than a general environmental classification. The transferable contribution is the experimental procedure; the Taxila percentiles, weights and tier thresholds require local reassessment at another property.

5.2 Interpreting persistence and disagreement

Giri, Giri Mosque and tombs, and Jaulian persisted across the structural scenarios, supporting a core inspection set. However, Bhallar illustrates why scenario frequencies must retain their definitions. It was seldom in the fixed structural top three but prominent after feature re-extraction and under spatial simulation. Combining those results into one unconditional confidence score would conceal the information most relevant to planning: Bhallar’s priority depends strongly on how its local terrain and spectral support are represented.

Joint intervals are wider than the certainty implied by a single ranking. They justify combining a persistent core set with additional visits targeted to particular mechanisms and representation-sensitive components. The exact crossing analysis supplies an interpretable decision margin: at 500 m, Giri’s leadership tolerates substantial domain-weight variation, whereas its 250 m ordering differs. This separates preference sensitivity from support sensitivity and identifies which assumption a manager must resolve. Access, cultural significance, repair history and specialist assessment would then guide the operational sequence.

5.3 What the predictive experiments establish

The seven-family comparison demonstrates useful geographic land-cover agreement, while the controls reveal where extra complexity adds little. Bands-only logistic regression remained competitive with the selected random forest, and temperature scaling worsened probability quality. Development selection and outer performance also diverged. Reporting these experiments together is more informative for model choice than selecting the largest test score: the appropriate model depends on spatial transfer, uncertainty and the intended output, not only point accuracy.

These findings are consistent with the importance of spatial evaluation and applicability domains [13, 15, 31]. Recent tabular foundation models offer additional comparative opportunities [43], but the present target remains consensus land cover. Class-stratified sampling, temporal mismatch and geographically concentrated water labels constrain interpretation. The previously examined outer stripe makes this a retrospective comparison, and neither it nor a more capable classifier provides independent monument-condition validation.

5.4 Climate interpretation and study limitations

Opposing product trends prevent a simple attribution of the landscape pattern to regional drying or warming. Acquisition-matched weather describes conditions relevant to the observations more directly, but five composite epochs provide limited temporal replication. Station measurements, soil observations and higher-frequency imagery would help distinguish climatic responses from management, seasonality and land-use processes [10, 44].

Point-centred neighbourhoods approximate landscape context rather than monument footprints; Saraikala lacks usable geometry. DEM routing cannot resolve every local obstruction, and percentile scores depend on the analysed inventory. Spatial simulations quantify stated perturbations without covering all label, sensor or conservation-model error. The next validation should therefore use independently assessed component-condition observations, record assessor agreement and evaluate components or periods excluded from design choices. Such evidence would test whether the proposed inspection strategy identifies actionable conservation problems.

6 Conclusions

CHIP links heterogeneous geospatial observations to inspection priorities through an explicit hierarchy and a coordinated experimental assessment. At Taxila, baseline comparisons and ablations showed that moisture-related spectral information and terrain pathways controlled important rank changes. Exact weight crossings quantified the preference intervals sustaining each leading component. Multiscale and structural scenarios identified a persistent core inspection set, while spatial reconstruction and joint simulations exposed additional, mechanism-dependent priorities. Acquisition-matched weather and product comparison constrained climatic interpretation. The separate spatial land-cover experiments demonstrated useful proxy agreement but also competitive simple controls and poor calibration transfer. Together, these results support a component-specific inspection programme with documented uncertainty, providing a practical basis for subsequent field validation and adaptation to other archaeological landscapes.

Data and code availability

The review package includes derived evidence, source manifests, executable notebooks, split records, configurations and result checksums. Original third-party products remain subject to provider terms. Identifying repository details are withheld from the anonymised manuscript and supplied separately to the editor. The workflow begins from documented derived data; source manifests describe upstream acquisition and preprocessing.

Declaration of generative AI and AI-assisted technologies

During preparation of this work, the authors used OpenAI ChatGPT/Codex to assist with code development and review, experiment execution, reproducibility checks, schematic preparation and manuscript drafting and editing. The authors are responsible for reviewing the code, numerical results, references and interpretations and for approving the final submission. The tool was not treated as an author or as scientific evidence.

References

- [1] Athos Agapiou, Dimitrios D. Alexakis, Vasiliki Lysandrou, Apostolos Sarris, Branka Cuca, Kyriacos Themistocleous, and Diofantos G. Hadjimitsis. Impact of urban sprawl to cultural

- heritage monuments: The case study of Paphos area in Cyprus. *Journal of Cultural Heritage*, 16(5):671–680, 2015. doi: 10.1016/j.culher.2014.12.006.
- [2] Athos Agapiou, Vasiliki Lysandrou, Kyriacos Themistocleous, and Diofantos G. Hadjimitsis. Risk assessment of cultural heritage sites clusters using satellite imagery and GIS: The case study of Paphos District, Cyprus. *Natural Hazards*, 83:5–20, 2016. doi: 10.1007/s11069-016-2211-6.
 - [3] Luigi Guerriero, Mario Di Napoli, Alessandro Novellino, Diego Di Martire, Ciro Rispoli, K. Lee, E. Bee, A. Harrison, and Domenico Calcaterra. Multi-hazard susceptibility assessment using analytic hierarchy process: The Derwent Valley Mills UNESCO World Heritage Site case study. *Journal of Cultural Heritage*, 55:339–345, 2022. doi: 10.1016/j.culher.2022.04.009.
 - [4] Muhammad Younis Khan, Federico Zaina, Zain ul Abedin, Shahina Tariq, and Jahangir Khan. Evaluation of risks to UNESCO World Heritage sites in Taxila, Pakistan using ground-based and satellite remote sensing techniques. *Journal of Cultural Heritage*, 55:195–209, 2022. doi: 10.1016/j.culher.2022.03.011.
 - [5] Najam us Saqib Zaheer Butt, Xinyuan Wang, Lei Luo, and Hammad Ul Hussan. Evaluating the impact of urban encroachment and land cover changes on world cultural heritage site taxila: A spatio-temporal analysis from 1990 to 2024. *Sustainability*, 17:1059, 2025. doi: 10.3390/su17031059.
 - [6] W.P. Megarry, T. Venkatachalam, J.C. Day, S. Jain, and S.F. Heron. Land use and land cover analysis of cultural world heritage to inform assessments of climate vulnerability. *Journal of Cultural Heritage*, 77:243–253, 2026. doi: 10.1016/j.culher.2025.11.008.
 - [7] Jacek Malczewski. GIS-based multicriteria decision analysis: A survey of the literature. *International Journal of Geographical Information Science*, 20(7):703–726, 2006. doi: 10.1080/13658810600661508.
 - [8] Bakhtiar Feizizadeh, Piotr Jankowski, and Thomas Blaschke. A GIS based spatially-explicit sensitivity and uncertainty analysis approach for multi-criteria decision analysis. *Computers & Geosciences*, 64:81–95, 2014. doi: 10.1016/j.cageo.2013.11.009.
 - [9] Risto Lahdelma and Pekka Salminen. Smaa-2: Stochastic multicriteria acceptability analysis for group decision making. *Operations Research*, 49:444–454, 2001. doi: 10.1287/opre.49.3.444.11220.
 - [10] Elena Sesana, Alexandre S. Gagnon, Chiara Ciantelli, JoAnn Cassar, and John J. Hughes. Climate change impacts on cultural heritage: A literature review. *WIREs Climate Change*, 12(4):e710, 2021. doi: 10.1002/wcc.710.
 - [11] Elena Sesana, Alexandre S. Gagnon, Alessandra Bonazza, and John J. Hughes. An integrated approach for assessing the vulnerability of World Heritage Sites to climate change impacts. *Journal of Cultural Heritage*, 41:211–224, 2020. doi: 10.1016/j.culher.2019.06.013.
 - [12] Kate Crowley, Rowan Jackson, Siona O’Connell, Dulma Karunarthna, Esti Anantasari, Arry Retnowati, and Dominique Niemand. Cultural heritage and risk assessments: Gaps, challenges, and future research directions for the inclusion of heritage within climate change adaptation and disaster management. *Climate Resilience and Sustainability*, 1(3):e45, 2022. doi: 10.1002/cli.245.
 - [13] Robert J. Hijmans. Cross-validation of species distribution models: Removing spatial sorting bias and calibration with a null model. *Ecology*, 93(3):679–688, 2012. doi: 10.1890/11-0826.1.

- [14] David R. Roberts, Volker Bahn, Simone Ciuti, Mark S. Boyce, Jane Elith, Gurutzeta Guillera-Arroita, et al. Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. *Ecography*, 40(8):913–929, 2017. doi: 10.1111/ecog.02881.
- [15] Hanna Meyer and Edzer Pebesma. Predicting into unknown space? estimating the area of applicability of spatial prediction models. *Methods in Ecology and Evolution*, 12(9): 1620–1633, 2021. doi: 10.1111/2041-210X.13650.
- [16] UNESCO World Heritage Centre. Taxila, 2026. URL <https://whc.unesco.org/en/list/139/>. Accessed 1 August 2026.
- [17] United States Geological Survey. Landsat collection 2 level-2 science products, 2026. URL <https://www.usgs.gov/landsat-missions/landsat-collection-2-level-2-science-products>. Accessed 1 August 2026.
- [18] David P. Roy, Valerii Kovalsky, Hankui K. Zhang, Eric F. Vermote, Lin Yan, S. S. Kumar, and Alexander Egorov. Characterization of Landsat-7 to Landsat-8 reflective wavelength and normalized difference vegetation index continuity. *Remote Sensing of Environment*, 185:57–70, 2016. doi: 10.1016/j.rse.2015.12.024.
- [19] Open-Meteo. Historical weather api, 2026. URL <https://open-meteo.com/en/docs/historical-weather-api>. Accessed 1 August 2026.
- [20] NASA POWER Project. Daily api documentation, 2026. URL <https://power.larc.nasa.gov/docs/services/api/temporal/daily/>. Accessed 1 August 2026.
- [21] European Space Agency. WorldCover data access, 2026. URL <https://esa-worldcover.org/en/data-access>. Accessed 1 August 2026.
- [22] Compton J. Tucker. Red and photographic infrared linear combinations for monitoring vegetation. *Remote Sensing of Environment*, 8(2):127–150, 1979. doi: 10.1016/0034-4257(79)90013-0.
- [23] Yong Zha, Jay Gao, and Shaoxiang Ni. Use of normalized difference built-up index in automatically mapping urban areas from TM imagery. *International Journal of Remote Sensing*, 24(3):583–594, 2003. doi: 10.1080/01431160304987.
- [24] Hanqiu Xu. Modification of normalised difference water index (NDWI) to enhance open water features in remotely sensed imagery. *International Journal of Remote Sensing*, 27(14):3025–3033, 2006. doi: 10.1080/01431160600589179.
- [25] Keith J. Beven and Michael J. Kirkby. A physically based, variable contributing area model of basin hydrology. *Hydrological Sciences Bulletin*, 24(1):43–69, 1979. doi: 10.1080/02626667909491834.
- [26] Ian D. Moore, Robert B. Grayson, and Andrew R. Ladson. Digital terrain modelling: A review of hydrological, geomorphological, and biological applications. *Hydrological Processes*, 5(1):3–30, 1991. doi: 10.1002/hyp.3360050103.
- [27] Y. Chen, J. Yu, and S. Khan. Spatial sensitivity analysis of multi-criteria weights in GIS-based land suitability evaluation. *Environmental Modelling & Software*, 25(12):1582–1591, 2010. doi: 10.1016/j.envsoft.2010.06.001.
- [28] Pranab Kumar Sen. Estimates of the regression coefficient based on Kendall’s tau. *Journal of the American Statistical Association*, 63(324):1379–1389, 1968. doi: 10.1080/01621459.1968.10480934.

- [29] Sheng Yue, Paul Pilon, Bob Phinney, and George Cavadias. The influence of autocorrelation on the ability to detect trend in hydrological series. *Hydrological Processes*, 16(9):1807–1829, 2002. doi: 10.1002/hyp.1095.
- [30] Yoav Benjamini and Yosef Hochberg. Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B*, 57(1):289–300, 1995. doi: 10.1111/j.2517-6161.1995.tb02031.x.
- [31] Hanna Meyer, Christoph Reudenbach, Tomislav Hengl, Marwan Katurji, and Thomas Nauss. Improving performance of spatio-temporal machine learning models using forward feature selection and target-oriented validation. *Environmental Modelling & Software*, 101: 1–9, 2018. doi: 10.1016/j.envsoft.2017.12.001.
- [32] Pontus Olofsson, Giles M. Foody, Martin Herold, Stephen V. Stehman, Curtis E. Woodcock, and Michael A. Wulder. Good practices for estimating area and assessing accuracy of land change. *Remote Sensing of Environment*, 148:42–57, 2014. doi: 10.1016/j.rse.2014.02.015.
- [33] Stephen V. Stehman and Giles M. Foody. Key issues in rigorous accuracy assessment of land cover products. *Remote Sensing of Environment*, 231:111199, 2019. doi: 10.1016/j.rse.2019.05.018.
- [34] Leo Breiman. Random forests. *Machine Learning*, 45:5–32, 2001. doi: 10.1023/A:1010933404324.
- [35] Fabian Pedregosa et al. Scikit-learn: Machine learning in python. *Journal of Machine Learning Research*, 12:2825–2830, 2011.
- [36] Tianqi Chen and Carlos Guestrin. Xgboost. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 785–794, 2016. doi: 10.1145/2939672.2939785.
- [37] Liudmila Prokhorenkova, Gleb Gusev, Aleksandr Vorobev, Anna Veronika Dorogush, and Andrey Gulin. Catboost: unbiased boosting with categorical features. In *Advances in Neural Information Processing Systems*, volume 31, 2018. URL <https://proceedings.neurips.cc/paper/2018/hash/14491b756b3a51daac41c24863285549-Abstract.html>.
- [38] Tilmann Gneiting and Adrian E. Raftery. Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477):359–378, 2007. doi: 10.1198/016214506000001437.
- [39] Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On calibration of modern neural networks. In *Proceedings of the 34th International Conference on Machine Learning*, volume 70 of *Proceedings of Machine Learning Research*, pages 1321–1330, 2017. URL <https://proceedings.mlr.press/v70/guo17a.html>.
- [40] Greg Wilson, Jennifer Bryan, Karen Cranston, Justin Kitzes, Lex Nederbragt, and Tracy K. Teal. Good enough practices in scientific computing. *PLOS Computational Biology*, 13(6): e1005510, 2017. doi: 10.1371/journal.pcbi.1005510.
- [41] Daniel Nüst, Carlos Granell, Barbara Hofer, Markus Konkol, Frank O. Ostermann, Rusne Sileryte, and Valentina Cerutti. Reproducible research and GIScience: An evaluation using AGILE conference papers. *PeerJ*, 6:e5072, 2018. doi: 10.7717/peerj.5072.
- [42] Xavier Romão and Chiara Bertolin. Risk protection for cultural heritage and historic centres: Current knowledge and further research needs. *International Journal of Disaster Risk Reduction*, 67:102652, 2022. doi: 10.1016/j.ijdrr.2021.102652.

- [43] Noah Hollmann, Samuel Müller, Lennart Purucker, Arjun Krishnakumar, Max Körfer, Shi Bin Hoo, Robin Tibor Schirrmeister, and Frank Hutter. Accurate predictions on small data with a tabular foundation model. *Nature*, 637:319–326, 2025. doi: 10.1038/s41586-024-08328-6.
- [44] Cathy Daly, Sandra Fatorić, Bethune Carmichael, Witiya Pittungnapoo, Olufemi Adetunji, Jørgen Hollesen, Masoud Nakhaei, and Alberto Herrera Diaz. Climate change adaptation policy and planning for cultural heritage in low- and middle-income countries. *Antiquity*, 96(390):1427–1442, 2022. doi: 10.15184/aqy.2022.114.